Research Paper Writing

Paper Anatomy, Section-by-Section Guidance,
and Academic Writing with AI

Research Workshop III

Dawei Chen

28 May 2026

Learning Objectives

- Understand the anatomy and logical flow of an empirical research paper
- Know what each section should accomplish and common mistakes to avoid
- Apply section-by-section guidelines to structure your own term paper
- Use AI tools responsibly to support — not replace — your academic writing
- Distinguish between effective AI-assisted writing and academic dishonesty

Three Example Papers

Each examines the impact of ChatGPT on Stack Overflow using natural experiments and DID designs.

P1

del Rio-Chanona et al. (2024)

Large language models reduce public knowledge sharing on online Q&A platforms. *PNAS Nexus*, 3(9), pgae400.

P2

Shan & Qiu (2025)

Examining the impact of generative AI on users' voluntary knowledge contribution. *Information Systems Research*, Article in Advance.

P3

Xue et al. (2026)

Can ChatGPT kill user-generated Q&A platforms? *Information Systems Research*, Article in Advance.

Anatomy of an Empirical Research Paper

Introduction	Literature Review	Hypothesis Dev.	Data & Method	Results	Discussion	Conclusion
Why does this matter?	*What do we already know?*	*What do we predict?*	*How do we test it?*	*What did we find?*	*What does it mean?*	*So what?*

Key Principles

Logical Flow Each section builds on the previous one. The reader should never feel lost.

Hourglass Shape Start broad (context), narrow to your specific question, then broaden to implications.

Every Section Has a Job No filler. If a paragraph doesn’t serve the section’s purpose, cut it.

Introduction: What It Should Accomplish

1	**Hook / Phenomenon**	Open with a compelling real-world phenomenon or striking fact.
2	**Research Gap**	Identify what prior literature missed, overlooked, or left untested.
3	**Research Question**	State your specific research question clearly and concisely.
4	**Preview of Approach**	Briefly describe data, setting, and empirical strategy (e.g., DID).
5	**Key Findings**	Summarize main results in 1–2 sentences before diving into details.
6	**Contributions**	Explain how your study advances the literature. What is new?

Introduction in Practice: How the Three Papers Do It

P1

Hook: LLMs substitute for human knowledge resources, threatening training data for future models.

Gap: Little evidence on whether LLM adoption reduces digital public goods.

RQ: *Does ChatGPT reduce posting activity on Stack Overflow?*

P2

Hook: Generative AI may complement or substitute voluntary knowledge contribution.

Gap: Impact of generative AI on answer behavior on Q&A platforms remains unexplored.

RQ: *Does generative AI use affect answer quantity, length, and readability?*

P3

Hook: LLMs offer instant answers, potentially displacing community Q&A platforms.

Gap: No empirical evidence on whether ChatGPT substitutes for question-asking.

RQ: *To what extent does ChatGPT substitute for Q&A platforms in volume and quality?*

Literature Review: Purpose & Structure

Purpose

- Establish the theoretical foundation for your research question
- Show what prior studies found — and what they missed
- Identify the specific gap your study fills
- Build the logical bridge from gap to hypotheses

Organize by Theme, Not by Paper

✗ Paper-by-paper summary

"Smith (2020) studied X. Jones (2021) studied Y."
→ *Reads like a list with no narrative.*

✓ Theme-based synthesis

"Studies converge on finding X (Smith, 2020; Jones, 2021), but the mechanism remains debated."

A literature review is not a book report. It is a structured argument that shows why your study is needed.

Literature Review: How the Three Papers Organize It

P1 **Themes:** LLMs and AI productivity | Digital public goods | Knowledge platform sustainability

Argues: *LLMs may crowd out the human-generated data they depend on.*

P2 **Themes:** Voluntary knowledge contribution | AI and human behavior | Cognitive load theory

Argues: *Generative AI may both help (learning) and hinder (cognitive load) contributions.*

P3 **Themes:** LLM impact on labor/content | Niche theory and media competition | UGC sustainability

Argues: *ChatGPT and Q&A platforms compete in overlapping niches with uneven substitution.*

Hypothesis Development: Building the Argument

A hypothesis is not a guess. It is a testable prediction derived from theory or prior evidence.

Theory / Logic

Start with a theoretical argument about the relationship.

→

Mechanism

Explain why this relationship would hold. What drives the effect?

→

Prediction

State the hypothesis formally as a testable statement.

Examples from the Papers

P1: ChatGPT substitutes for Stack Overflow → 25% decline in posts (lower bound estimate)

P2: Learning → more answers; Cognitive load at high intensity → fewer answers

P3: Uneven substitution → simple questions migrate; remaining quality rises

Data & Methodology: What to Cover

Research Setting	Describe the platform or context. Why is it a good setting?
Data Source & Collection	Where, when, how many observations, key variables?
Identification Strategy	How do you establish causality? Natural experiment + DID?
Variable Definitions	DV and IV defined precisely. Include a summary table.
Model Specification	Regression equation, fixed effects, controls, and rationale.

Comparing Research Designs Across the Three Papers

	del Rio-Chanona et al. (2024)	Shan & Qiu (2025)	Xue et al. (2026)
Natural Experiment	ChatGPT launch (Nov 30, 2022)	ChatGPT launch (Nov 30, 2022)	ChatGPT launch (Nov 30, 2022)
Treatment	Stack Overflow (English)	Users who adopted ChatGPT	Stack Overflow (2022–23)
Control	Russian SO, Chinese Segmentfault, Math SE	Non-adopting users	Same months in 2021–22
Method	Difference-in-Differences (DID)	Difference-in-Differences (DID)	Difference-in-Differences (DID)
Data Source	Stack Exchange archive + Segmentfault scrape	Stack Overflow (Sept–Dec 2022)	Stack Overflow dumps (quarterly)
Key DV	Post count, votes	Answer count, length, readability	Question count, score, SMOG complexity

All three papers share the same core design: a natural experiment analyzed with DID.

Results: Presenting Your Findings

Descriptive Statistics

Summary table of means, SDs, min/max. Orients the reader.

Visualizations

Trend lines, parallel trends plots, event study graphs. A figure can communicate more than a table.

Main Results

Core regression results: coefficient, significance, plain-language interpretation.

Robustness Checks

Alternative specifications, time windows, placebo tests. Builds credibility.

Report what you found, even if it contradicts your hypotheses. Null findings are legitimate.

Results: Key Findings Across the Three Papers

P1

- **Main findings:** Stack Overflow posting declined 25% relative to counterfactuals. No change in post quality. Decline spans all experience levels; popular languages most affected.
- **Robustness:** *Platform-specific trends, alternative time windows, generalized DID, developer survey corroboration.*

P2

- **Main findings:** ChatGPT adopters generated more answers (+16.77%), shorter and more readable. Cumulative use enhances learning; intensive use increases cognitive load.
- **Robustness:** *Event study design, propensity score matching, alternative treatment windows.*

P3

- **Main findings:** 14.09% average reduction in questions; 27.88% by May 2023. Remaining questions show higher quality and complexity. Uneven substitution: mid/low-quality questions displaced most.
- **Robustness:** *Parallel trend verification, alternative control platforms, heterogeneity by user type.*

Discussion: Interpreting Your Results

Step back from the numbers and explain what they mean.

Interpret Results	What do coefficients mean practically? Connect back to hypotheses.
Relate to Literature	Confirm, extend, or contradict prior work? Show your contribution.
Theoretical Implications	Does it support a mechanism? Challenge an assumption?
Practical Implications	What should managers, policymakers, or designers do differently?
Limitations & Future Work	Threats to validity? What would stronger data or methods reveal?

Conclusion: Closing the Loop

Typically 0.5–1 page. Short, direct, and forward-looking.

- Restate the research question and summarize main findings (1–2 sentences)
- Highlight the most important contribution of the paper
- Briefly mention practical takeaways for practitioners or policymakers
- End with a forward-looking statement about future research directions

Avoid: introducing new data or arguments; restating every result; overgeneralizing beyond what the data supports; ending on a vague, generic statement.

A Practical Workflow for AI-Assisted Academic Writing

1. **Think First**
   Outline your argument, key points, and section structure before using AI.

2. **Draft with AI**
   Generate a rough draft from your outline. Be specific: section, argument, data, length.

3. **Critically Read**
   Is it accurate? Does it match your data? Delete anything unsupported.

4. **Revise & Rewrite**
   Rewrite in your own voice. If you can’t explain a sentence, you don’t understand it.

5. **Verify & Document**
   Check all citations (many are fabricated). Verify facts. Document AI usage.

Your Term Paper: Section-by-Section Checklist

Use this as a self-assessment before submitting.

Section	Include	✓
Introduction	Hook + gap + RQ + preview + findings + contributions	☐
Literature Review	Organized by themes, not papers; ends with clear gap	☐
Hypotheses	Theory → mechanism → testable prediction; formally stated	☐
Data & Method	Source, collection, variables, model specification, identification	☐
Results	Descriptive stats, main results, visuals, robustness checks	☐
Discussion	Interpretation, comparison to literature, implications, limitations	☐
Conclusion	Summary of findings, contribution, future directions	☐
References	All cited; all real; consistent format (APA or journal style)	☐

Key Reminders & Timeline

- **Jun 2 (Tue)** Research Workshop IV — No lecture, consultation
- **Jun 4 (Thu)** Final Project Presentations — Session 1
- **Jun 9 (Tue)** Final Project Presentations — Session 2
- **Jun 10, 23:59** Term Paper Due (Beijing Time)

Presentation Guidelines

- 6 min presentation + 2 min Q&A, 15% of final grade
- Upload slides to Canvas (Final Presentation) before 9:30 AM on your presentation day
- Attendance at both sessions is mandatory, regardless of your presentation date
- See "*final_project_instructions_v5.0.pdf*" in Canvas for detailed guidelines

Course Evaluation

Spring 2026 Course Evaluations are now open!

Deadline: **June 3, 2026, 23:59** (China time)

Link: https://kean.campuslabs.com/eval-home/direct/0072436

- Check your @kean.edu email (including spam) for the survey
- Only takes several minutes; works on smartphones
- You may complete it outside the classroom

Your feedback is confidential and helps improve the course.

Q&A

dchen@kean.edu | CBPM B214

Office hours: Tue & Thu 11:15–12:30 | Wed 14:00–16:30